

K-Nearest Neighbors: KNN Classification versus KNN Regression

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Abstract

K-Nearest Neighbors (KNN) is a non-parametric supervised learning method that can be used for both classification and regression. Although KNN classification and KNN regression share the same fundamental procedure of identifying observations that are close to a new observation, they differ in the type of response variable and in the way the neighboring observations are used to produce predictions. KNN classification is appropriate when the response is qualitative or categorical and generally uses a majority-vote rule, whereas KNN regression is appropriate when the response is quantitative and generally uses the average response of the nearest neighbors. This paper explains the conceptual and mathematical differences between the two approaches, discusses the role of K , explains why predictor standardization is important, and considers the advantages, disadvantages, and connection between KNN and the curse of dimensionality. The discussion is based primarily on the statistical-learning framework presented by James et al. [1].

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1 Introduction

K-Nearest Neighbors (KNN) is a non-parametric supervised learning method that can be used for both classification and regression. The fundamental idea is the same in both cases: when making a prediction for a new observation, KNN identifies the K training observations that are closest to it according to a chosen distance measure. However, the two methods differ in what they predict and how they combine the outcomes of the nearest neighbors.

KNN classification predicts a qualitative or categorical response, whereas KNN regression predicts a quantitative or numerical response. This distinction is central to the statistical-learning framework presented by James et al. [1]. In both cases, the method does not require the researcher to specify a particular global functional form linking the predictors to the response. Instead, KNN relies on the assumption that observations that are close to one another in predictor space are likely to have similar response values.

The distinction between classification and regression is therefore particularly important when selecting an appropriate statistical-learning method. Classification is concerned with predicting a class or category, such as whether a customer will default on a loan, while regression is concerned with predicting a numerical quantity, such as the amount a customer will spend. This paper provides a detailed comparison of KNN classification and KNN regression.

2 What is KNN?

Suppose we have a training dataset containing observations

$$(X_1, Y_1), (X_2, Y_2), \dots, (X_n, Y_n), \quad (1)$$

where X_i represents the predictor variables and Y_i represents the response variable.

For a new observation x_0 , KNN follows three basic steps:

1. Measure the distance between x_0 and every observation in the training dataset.
2. Identify the K observations that are closest to x_0 .
3. Use the response values of those K neighbors to make a prediction.

For numerical predictors, Euclidean distance is commonly used. The distance between a new observation x_0 and a training observation x_i can be written as

$$d(x_0, x_i) = \sqrt{\sum_{j=1}^p (x_{0j} - x_{ij})^2}, \quad (2)$$

where:

- p is the number of predictor variables;
- x_{0j} is the value of predictor j for the new observation;
- x_{ij} is the value of predictor j for training observation i .

The important point is that KNN classification and KNN regression use the same basic neighbor-searching mechanism but make different types of predictions.

3 KNN Classification

3.1 Definition

KNN classification is used when the response variable Y is qualitative or categorical. Examples include:

- Disease versus no disease;
- Default versus no default;
- Success versus failure;
- Spam versus not spam;
- Low, medium, or high risk.

The objective is to determine which class a new observation belongs to.

3.2 How KNN Classification Works

Suppose that the possible classes are

$$A, \quad B, \quad C. \tag{3}$$

For a new observation x_0 , KNN identifies its K nearest neighbors and examines their class labels. The predicted class is generally the class receiving the majority of votes.

Mathematically, the predicted class can be expressed as

$$\hat{Y}_0 = \arg \max_c \sum_{i \in N_0} I(Y_i = c), \tag{4}$$

where:

- N_0 is the set of the K nearest neighbors of x_0 ;
- c represents a possible class;
- $I(Y_i = c)$ is an indicator function that equals 1 if observation i belongs to class c and 0 otherwise.

3.3 Example

Suppose $K = 5$, and the five nearest observations have the following labels:

Table 1: Example of KNN Classification

Neighbor	Class
1	Yes
2	Yes
3	No
4	Yes
5	No

There are three observations classified as “Yes” and two classified as “No”. Therefore,

$$\hat{Y}_0 = \text{Yes.} \quad (5)$$

The new observation is therefore assigned to the “Yes” class because Yes receives the majority vote.

4 KNN Regression

4.1 Definition

KNN regression is used when the response variable Y is quantitative or numerical. Examples include:

- House price;
- Salary;
- Temperature;
- Sales revenue;
- Miles per gallon (MPG);
- Blood pressure.

Instead of predicting a class, KNN regression predicts a numerical value.

4.2 How KNN Regression Works

KNN regression identifies the K nearest observations and calculates the average of their response values.

The basic KNN regression estimator is

$$\hat{f}(x_0) = \frac{1}{K} \sum_{i \in N_0} Y_i. \quad (6)$$

Thus, unlike classification, where the neighbors vote, regression averages the numerical responses of the nearest neighbors.

4.3 Example

Suppose the five nearest neighbors have the following house prices:

\$200,000, \$220,000, \$210,000, \$250,000, \$220,000.

The KNN regression prediction is

$$\hat{Y}_0 = \frac{200000 + 220000 + 210000 + 250000 + 220000}{5} \quad (7)$$

$$= 220000. \quad (8)$$

Therefore,

$$\boxed{\hat{Y}_0 = \$220,000}. \quad (9)$$

5 The Main Difference: Voting versus Averaging

The most important distinction between KNN classification and KNN regression is the way the nearest neighbors are combined.

5.1 KNN Classification

In KNN classification, the neighbors vote:

$$\boxed{\text{Prediction} = \text{Majority Class}} \quad (10)$$

5.2 KNN Regression

In KNN regression, the numerical responses of the neighbors are averaged:

$$\boxed{\text{Prediction} = \text{Average Numerical Response}} \quad (11)$$

Consider five nearest observations with numerical response values

$$10, \quad 12, \quad 13, \quad 20, \quad 25.$$

If the observations represent a regression problem, KNN regression calculates

$$\hat{Y}_0 = \frac{10 + 12 + 13 + 20 + 25}{5} = 16. \quad (12)$$

Therefore, the predicted numerical response is

$$\boxed{16}. \quad (13)$$

If the observations instead represent a classification problem, their class labels would be used to determine the majority class.

6 Comparison of KNN Classification and KNN Regression

Table 2 summarizes the major differences between KNN classification and KNN regression.

Table 2: Comparison of KNN Classification and KNN Regression

Feature	KNN Classification	KNN Regression
Type of response	Qualitative/categorical	Quantitative/numerical
Purpose	Predict a class	Predict a numerical value
Example	Disease / No disease	House price
Neighbor selection	K nearest observations	K nearest observations
Prediction method	Majority vote	Average
Output	Class label or class probabilities	Numerical value
Typical loss function	Classification error	Squared error or related regression loss
Example prediction	“Yes”	\$250,000
Effect of K	Controls smoothness of the decision boundary	Controls smoothness of the fitted regression function

7 The Role of K

The value of K is important in both KNN classification and KNN regression.

7.1 Small K

Suppose

$$K = 1. \tag{14}$$

The prediction depends entirely on the single closest observation. This produces a highly flexible model because a small movement in the predictor space can cause the nearest observation to change.

Consequently, small values of K can result in relatively low bias but high variance.

For KNN classification, the class of the nearest observation determines the prediction.

For KNN regression, the numerical response of the nearest observation determines the prediction.

7.2 Large K

When K is large, the prediction uses many observations, producing a smoother model. This generally reduces variance but can increase bias.

For example, if

$$K = n, \tag{15}$$

then every training observation becomes a neighbor.

For KNN regression,

$$\hat{Y}_0 = \frac{1}{n} \sum_{i=1}^n Y_i. \tag{16}$$

Thus, every prediction becomes the overall mean of the response variable.

For KNN classification, every prediction tends toward the most common class in the training data.

8 A Practical Example

Consider a bank that wants to predict whether a new customer will default on a loan.

The predictors may include:

- Income;

- Age;
- Loan amount;
- Credit score.

The response is

$$Y = \begin{cases} \text{Default,} \\ \text{No Default.} \end{cases} \quad (17)$$

Because the response is categorical, this is a classification problem. Suppose the five nearest customers have the following outcomes:

$$\text{Default, No Default, Default, Default, No Default.} \quad (18)$$

There are three “Default” observations and two “No Default” observations. Therefore,

$$\boxed{\hat{Y}_0 = \text{Default}}. \quad (19)$$

Now suppose that the bank wants to predict the amount of money the customer will spend next month. The response is now numerical:

$$Y = \text{Monthly Spending.} \quad (20)$$

Suppose the five nearest customers spent

$$\$400, \quad \$500, \quad \$450, \quad \$550, \quad \$600.$$

KNN regression predicts

$$\hat{Y}_0 = \frac{400 + 500 + 450 + 550 + 600}{5} \quad (21)$$

$$= 500. \quad (22)$$

Therefore,

$$\boxed{\hat{Y}_0 = \$500}. \quad (23)$$

The neighbor-selection process is the same, but the way the outcomes are combined is different.

9 Why Standardization is Important

An important practical issue for both KNN classification and KNN regression is the measurement scale of the predictors.

Suppose that two predictors are:

- Age: 18–80 years;
- Income: \$20,000–\$200,000.

If Euclidean distance is used directly, income can dominate the distance calculation because it is measured on a much larger numerical scale.

For two observations, the distance might be represented as

$$d(x_1, x_2) = \sqrt{(\text{Age}_1 - \text{Age}_2)^2 + (\text{Income}_1 - \text{Income}_2)^2}. \quad (24)$$

The income difference could therefore overwhelm the age difference.

To reduce this problem, predictors are often standardized using

$$Z_j = \frac{X_j - \bar{X}_j}{s_j}, \quad (25)$$

where $\bar{X}_j$ is the sample mean and s_j is the sample standard deviation of predictor j . After standardization, each predictor has approximately

$$\text{Mean} = 0, \quad \text{Standard Deviation} = 1. \quad (26)$$

Standardization is important because KNN relies directly on distances between observations.

10 Advantages and Disadvantages

10.1 KNN Classification

Advantages

1. KNN classification is simple to understand and implement.
2. It does not require a strong parametric assumption about the relationship between predictors and the response.
3. It can model complicated and nonlinear decision boundaries.
4. It can naturally handle multiple classes.

Disadvantages

1. Prediction can be computationally expensive for large datasets.
2. Performance can deteriorate when there are many predictors.
3. KNN is sensitive to the scale of predictors.
4. Results are sensitive to the choice of K .
5. Irrelevant predictors can negatively affect distance calculations.

10.2 KNN Regression

Advantages

1. KNN regression is simple and intuitive.
2. It can model nonlinear relationships.
3. It makes relatively few assumptions about the functional form of the regression relationship.
4. It is useful when nearby observations are expected to have similar response values.

Disadvantages

1. It is sensitive to predictor scaling.
2. It can be affected by irrelevant variables.
3. It can perform poorly in high-dimensional settings.
4. It requires the training data to be available when making predictions.
5. The choice of K is important.
6. Predictions can be computationally expensive for large datasets.

11 Connection with the Curse of Dimensionality

KNN methods are particularly affected by the *curse of dimensionality*. As the number of predictors p increases, observations tend to become increasingly far apart in predictor space.

Consequently, the concept of a “nearest” neighbor becomes less informative.

For example, suppose that house prices are predicted using only two predictors:

$$p = 2. \tag{27}$$

The predictors might be:

- House size;
- Number of bedrooms.

Finding nearby houses in a two-dimensional predictor space may be relatively straightforward.

However, suppose that 100 predictors are used:

$$p = 100. \tag{28}$$

Finding observations that are genuinely similar across all 100 dimensions becomes much more difficult. This can substantially reduce the effectiveness of KNN.

Therefore, KNN often works best when the number of relevant predictors is relatively small.

12 KNN versus Linear Regression

It is also useful to distinguish KNN regression from ordinary linear regression.

12.1 Linear Regression

Linear regression assumes a particular functional form. For example,

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon. \tag{29}$$

The parameters β_0 , β_1 , and β_2 are estimated from the training data and are then used to make predictions.

12.2 KNN Regression

KNN regression does not specify a global linear functional form. Instead, it essentially asks:

Find observations that are similar to this new observation and use their response values to estimate the response.

Consequently, KNN regression can capture nonlinear relationships without explicitly specifying a polynomial or other global functional form.

However, KNN generally requires sufficient observations near the point where a prediction is being made.

13 Summary

The fundamental distinction between KNN classification and KNN regression can be summarized as follows:

$$\begin{array}{lcl} \text{KNN Classification} & \longrightarrow & \text{Categorical response} \longrightarrow \text{Majority vote} \\ \text{KNN Regression} & \longrightarrow & \text{Numerical response} \longrightarrow \text{Average} \end{array} \quad (30)$$

In KNN classification, the K nearest observations vote on the class of the new observation. The class receiving the greatest number of votes becomes the prediction.

In KNN regression, the K nearest observations contribute their numerical response values, which are typically averaged to produce the prediction.

Therefore, the fundamental neighbor-searching procedure is the same, but the response type and prediction rule are different. Both methods are flexible and non-parametric, but both are also sensitive to the choice of K , predictor scaling, irrelevant variables, and the number of dimensions.

The choice between KNN classification and KNN regression should therefore be based first on the nature of the response variable. If the response is categorical, KNN classification is appropriate; if the response is quantitative, KNN regression is appropriate. In both cases, careful consideration should also be given to the choice of K , predictor scaling, and dimensionality.

14 Conclusion

K-Nearest Neighbors provides a flexible framework for supervised learning in which predictions are based on observations that are close to a new observation in predictor space. The key distinction between the classification and regression versions lies in the nature of the response variable and the rule used to combine the nearest neighbors.

KNN classification is designed for categorical outcomes and uses majority voting, while KNN regression is designed for numerical outcomes and generally uses averaging. Although both approaches are conceptually simple and capable of representing nonlinear relationships, their performance depends strongly on the quality and scale of the predictors, the choice of K , and the dimensionality of the predictor space.

Overall, KNN classification and KNN regression should not be viewed as completely different algorithms. Rather, they are two applications of the same nearest-neighbor principle to two different types of supervised-learning problems.

References

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